

We use random variables (RVs) for representing our knowledge about certain variables of systems.

We use Probability Density Functions (PDF) for describing the statistical properties of RVs (i.e., our “belief” about that variable).

When a RV ξ is statistically described by a PDF $f(\xi)$ we usually express that fact as: $\xi \sim f(\xi)$

In the case of a system's states.

“We can express our belief about those states
through a PDF”

From our perspective, that hidden state is a RV,
because we do not know its exact value,
thus, we have a “belief “about it,
which we express through a PDF

We obtain that PDF based on the sources of information that are
available to us,

Statistical representation (PDFs)

Given a **scalar RV** ξ which is statistically described by a PDF $p_\xi(\xi)$; that PDF must satisfy the following:

$$\xi \sim p_\xi(\xi)$$

$$p_\xi(\cdot) : \mathbb{R}^1 \rightarrow \mathbb{R}^1$$

$$p_\xi(\xi) \geq 0, \quad \forall \xi \in \mathbb{R}^1$$

$$\int_{-\infty}^{\infty} p_\xi(\xi) \cdot d\xi = 1$$

$$\int_{\text{Domain of } \xi} p_\xi(\xi) \cdot d\xi = 1$$

statistical representation

If, given a PDF $p_{\xi}(\xi)$ which models the statistical properties of the RV ξ (i. e. $\xi \sim f(\xi)$), we want to know the probability of ξ having its value contained in the interval **$[a, b]$** , then that probability is provided by the following operation,

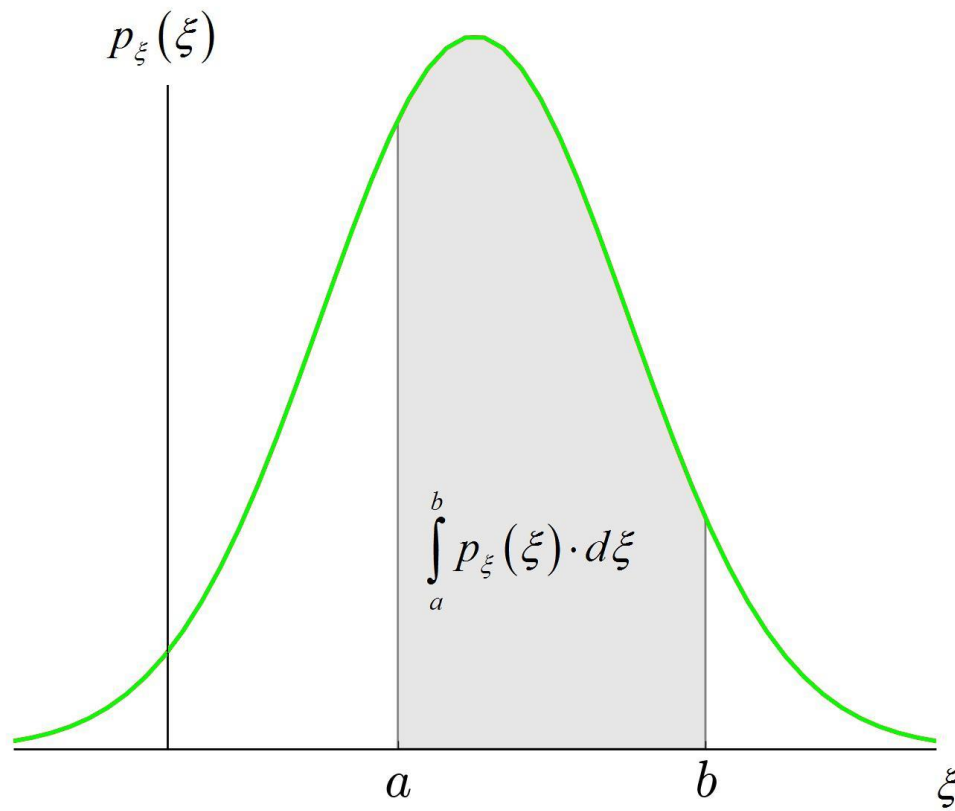
$$p(\xi \in [a, b]) = \int_a^b p_{\xi}(\xi) \cdot d\xi$$

Example:

“The probability that the temperature of this room, tomorrow at 2PM will be in the range $25^{\circ}\text{C} < T < 30^{\circ}\text{C}$, is 0.3 (30%).”

It is the probability of the RV having its actual value in that region $[a, b]$

$$p(\xi \in [a, b]) = \int_a^b p_\xi(\xi) \cdot d\xi$$



Multivariate PDFs

PDFs are used for describing RVs.

Estimates of the states of a system can be treated as being RVs

//the states do usually have value (it is not a RV) but because we do not exactly know that value, we have a PDF for representing our knowledge about the value the states may have. So, it is a RV from our perspective, due to our uncertainty.

We can express our belief about those estimates through a PDF.

An estimate of a state vector is usually a multidimensional variable.

In general, a multidimensional RV can be seen as a vector whose components are scalars, each of them being a scalar RV.)

Those multiple RVs need to be jointly described.

A PDF that describes multiple scalar RVs is said to be a multivariate PDF.

This PDF is called the JOINT PDF of those scalar RVs.

Multivariate PDFs

$$\mathbf{x} = \begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix}^T \in \mathbb{R}^n$$

$$\mathbf{x} \sim p_{\mathbf{x}}(\mathbf{x})$$

$$p_{\mathbf{x}}(.) : \mathbb{R}^n \rightarrow \mathbb{R}^1$$

We can also express it in this way:

$$p_{x_1, x_2, \dots, x_n} (x_1, x_2, \dots, x_n)$$

Multivariate PDFs

A Multivariate PDF does also satisfy:

$$p_{x_1, x_2, \dots, x_n}(x_1, x_2, \dots, x_n) \geq 0, \quad \forall (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$$

$$\int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} p_{x_1, x_2, \dots, x_n}(x_1, x_2, \dots, x_n) \cdot dx_1 \cdot dx_2 \cdot \dots \cdot dx_n = 1$$

$$p_{\mathbf{X}}(\mathbf{X}) \geq 0, \quad \forall \mathbf{X} \in \mathbb{R}^n$$

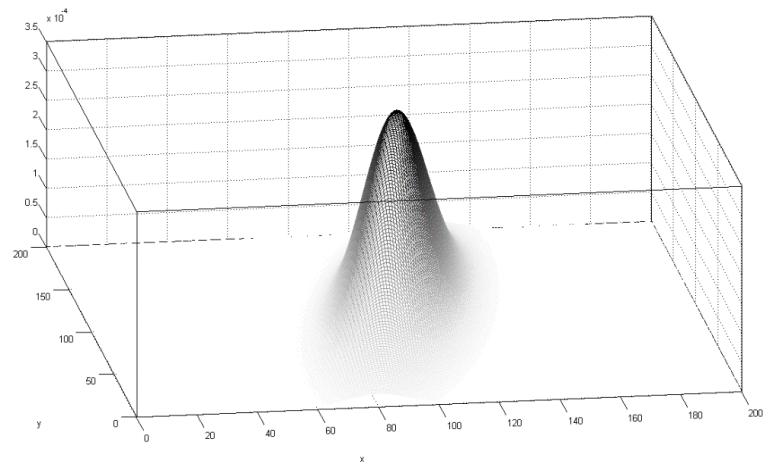
$$\int_{\text{domain of } \mathbf{X}} p_{\mathbf{X}}(\mathbf{X}) \cdot d\mathbf{X} = 1$$

Multivariate PDFs

Example?

PDF about a vehicle position, in 2D.

$$p_{x,y}(x, y)$$



Case: PDF about the vehicle pose, in 2D (so, state vector is 3D).

$$p_{x,y,\varphi}(x,y,\varphi)$$

What you are assuming as a “deterministic” variable in Project 1 would need to be represented as a RV and expressed via a PDF.

How to obtain that PDF?

That will be our task/goal in lecture 4.

Multivariate PDFs

$$p(\mathbf{x} \in \Omega) = \int_{\Omega} p_{\mathbf{x}}(\mathbf{x}) \cdot d\mathbf{x}$$

(This is a definition/concept.

We do not usually perform this calculation)

Definition: Marginal PDF

Even if we have the joint PDF about a set of RVs, we may ask for the description of an individual variable or for certain subset of the variables. In that case we can *marginalize* the PDF.

For the “easy” case of having the joint PDF of 2 scalar RVs, the mathematical definition is shown by the following expressions:

I got $p_{x,y}(x,y)$ but I just need $p_x(x)$; how do I get it?

$$p_x(x) = \int_{-\infty}^{+\infty} p_{x,y}(x,y) \cdot dy, \quad p_y(y) = \int_{-\infty}^{+\infty} p_{x,y}(x,y) \cdot dx$$

...Marginal PDF

Higher dimensions can be considered as well; here we express it in a general way, using vectors (and “volume” differentials).

Given the joint PDF $p_{\mathbf{X},\mathbf{Y}}(\mathbf{X}, \mathbf{Y})$

We want to obtain the marginal PDF $p_{\mathbf{X}}(\mathbf{X})$

$$p_{\mathbf{X}}(\mathbf{X}) = \int_{\substack{\text{domain} \\ \text{of } \mathbf{Y}}} p_{\mathbf{X},\mathbf{Y}}(\mathbf{X}, \mathbf{Y}) \cdot d\mathbf{Y}$$

...Marginal PDF

Case in which we use scalar RVs, and have the joint PDF $p_{x,y,z}(x, y, z)$

But we have interest only on x .

So, we can obtain the PDF $p_x(x)$

By marginalization:

$$p_x(x) = \int_{\substack{\text{domain} \\ \text{of } (y,z)}} p_{x,y,z}(x, y, z) \cdot dy \cdot dz$$

Definition: Conditional PDF

Given our joint belief about the RVs $(\mathbf{x}, \mathbf{y})$, $p_{\mathbf{x},\mathbf{y}}(\mathbf{x}, \mathbf{y})$, what would be our belief about $\mathbf{x}$ if we knew that the actual value of $\mathbf{y}$ is $\mathbf{y}_0$?

We express it as follows:

$$p_{\mathbf{x}|\mathbf{y}}(\mathbf{x} | \mathbf{y} = \mathbf{y}_0)$$

Note that this PDF is about the RV $\mathbf{x}$

“Probability of $\mathbf{x}$ provided that $\mathbf{y}=\mathbf{y}_0$ ” ($\leftarrow$ conditional)

Conditional PDF

In general, given a joint PDF about the RVs $(\mathbf{x}, \mathbf{y})$, $p_{\mathbf{x},\mathbf{y}}(\mathbf{x}, \mathbf{y})$, we can obtain the conditional PDF $p_{\mathbf{x}|\mathbf{y}}(\mathbf{x}|\mathbf{y})$ as follows:

$$p_{\mathbf{x}|\mathbf{y}}(\mathbf{x} | \mathbf{y}) = \frac{p_{\mathbf{x},\mathbf{y}}(\mathbf{x}, \mathbf{y})}{p_{\mathbf{y}}(\mathbf{y})}$$

Similarly

$$p_{\mathbf{y}|\mathbf{x}}(\mathbf{y} | \mathbf{x}) = \frac{p_{\mathbf{x},\mathbf{y}}(\mathbf{x}, \mathbf{y})}{p_{\mathbf{x}}(\mathbf{x})}$$

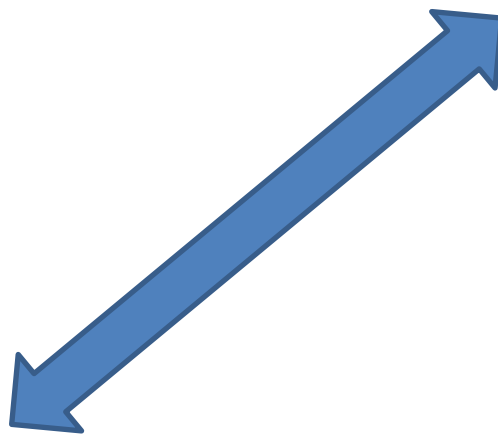
$$p_{\mathbf{x},\mathbf{y}}(\mathbf{x}, \mathbf{y})$$

$$p_{\mathbf{x}}(\mathbf{x})$$

$$p_{\mathbf{y}}(\mathbf{y})$$

$$p_{\mathbf{x}|\mathbf{y}}(\mathbf{x} | \mathbf{y})$$

$$p_{\mathbf{y}|\mathbf{x}}(\mathbf{y} | \mathbf{x})$$



Definition: statistical independence

We say that two RVs, $(\mathbf{x}, \mathbf{y})$ (which are statistically described by the joint PDF, $p_{\mathbf{x}, \mathbf{y}}(\mathbf{x}, \mathbf{y})$), are **independent**, if their joint PDF can be factorized as follows,

$$(\mathbf{x}, \mathbf{y}) \sim p_{\mathbf{x}, \mathbf{y}}(\mathbf{x}, \mathbf{y}) = p_{\mathbf{x}}(\mathbf{x}) \cdot p_{\mathbf{y}}(\mathbf{y})$$

(independent RVs..)

That condition implies that their conditional PDFs are equal to their marginal ones, i.e.

$$p_{\mathbf{x}|\mathbf{y}}(\mathbf{x} | \mathbf{y}) = \frac{p_{\mathbf{x},\mathbf{y}}(\mathbf{x}, \mathbf{y})}{p_{\mathbf{y}}(\mathbf{y})} = p_{\mathbf{x}}(\mathbf{x})$$

$$p_{\mathbf{y}|\mathbf{x}}(\mathbf{y} | \mathbf{x}) = \frac{p_{\mathbf{x},\mathbf{y}}(\mathbf{x}, \mathbf{y})}{p_{\mathbf{x}}(\mathbf{x})} = p_{\mathbf{y}}(\mathbf{y})$$

(independent RVs..)

Any knowledge we may have about Y would not affect our “belief” about X .

A situation such this may be beneficial or not, depending on the cases in which the PDF is being used.

The Gaussian Representation

As in many other areas of engineering and sciences, where the models are simplified by using tractable approximations, we need, in certain cases, to approximate the PDF functions.

The Gaussian Distribution

Consider a scalar RV x . If it is said to be a Gaussian RV, then its PDF will have the following structure

$$p_x(x) = c \cdot e^{-\frac{(x-\hat{x})^2}{2 \cdot \sigma_x^2}}$$

$$c = \frac{1}{\sigma_x \cdot \sqrt{2 \cdot \pi}} : \begin{array}{l} \text{normalizing factor} \\ \text{(so that the function has "volume=1")} \end{array}$$

The Gaussian Representation

It has two parameters

$$p_x(x) = c \cdot e^{-\frac{(x - \hat{x})^2}{2 \cdot \sigma_x^2}}$$

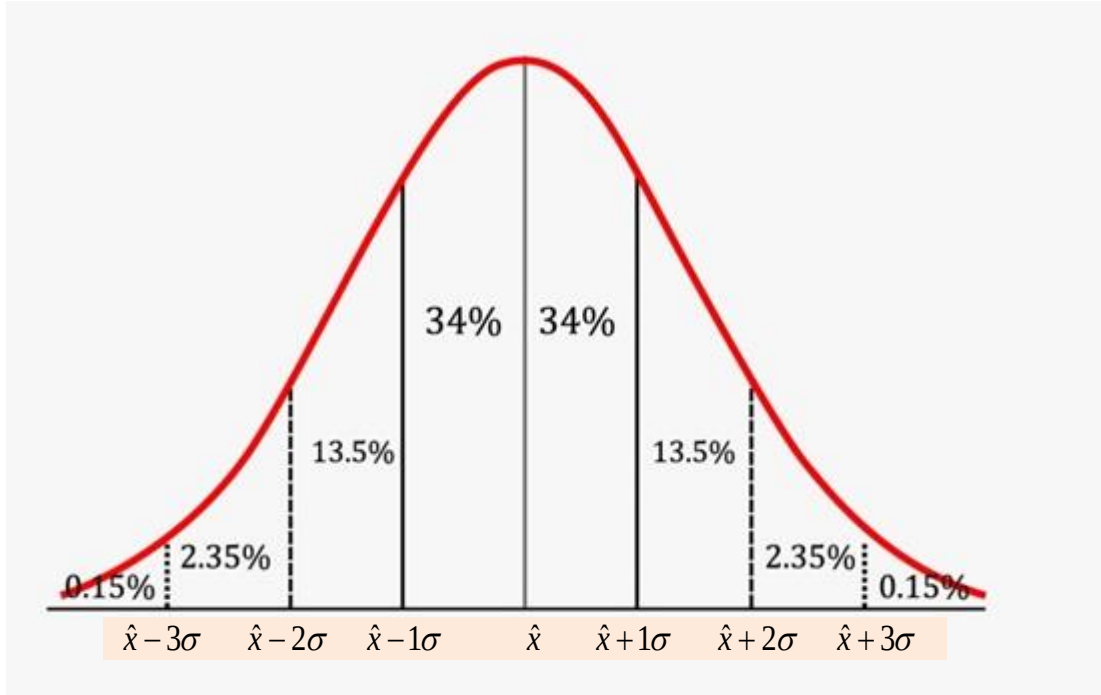
$\hat{x}$: expected value

σ_x^2 : variance

(σ_x : standard deviation)

$$c = \frac{1}{\sigma_x \cdot \sqrt{2 \cdot \pi}} : \text{normalizing factor (so that the function has "volume=1")}$$

The Gaussian Representation



$$p_x(x) = c \cdot e^{-\frac{(x - \hat{x})^2}{2 \cdot \sigma^2}}$$

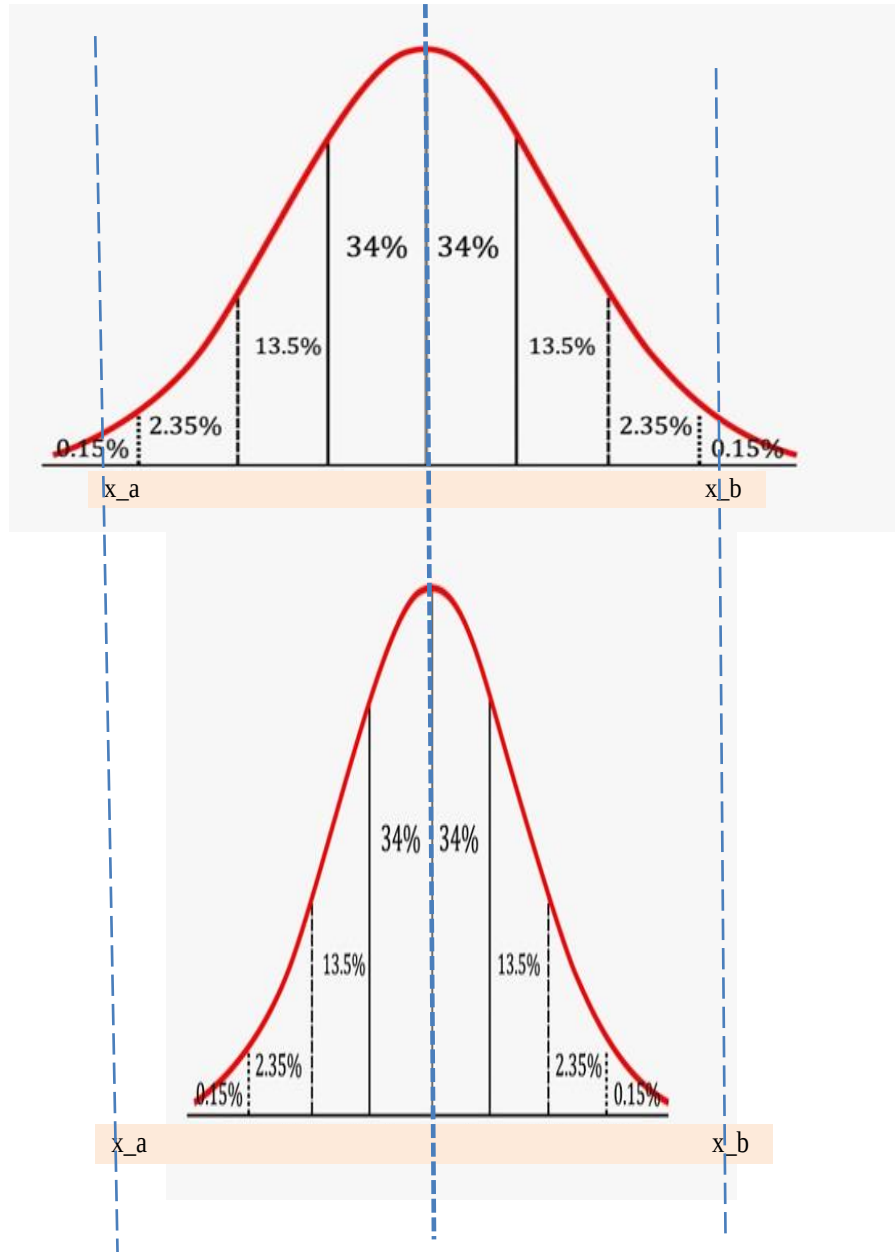
$\hat{x}$: expected value (a.k.a. "mean")

σ^2 : variance

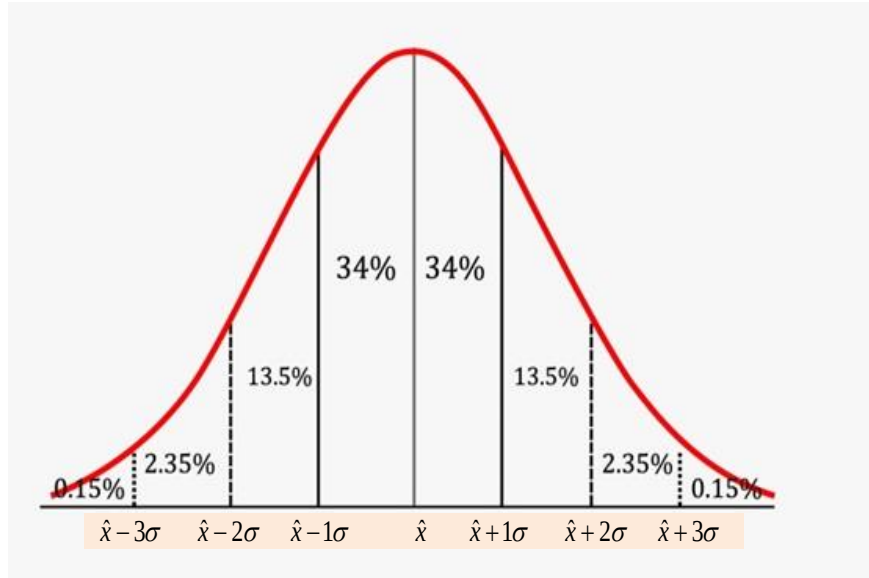
c : normalizing constant

σ : standard deviation (what is telling us σ ?)

(confidence?)



The Gaussian Representation



$$p_x(x) = c \cdot e^{-\frac{(x - \hat{x})^2}{2 \cdot \sigma^2}}$$

σ : what would mean $\sigma \rightarrow 0$?

$$p_x(x) \rightarrow \delta(x - \hat{x}) \quad (\text{Dirac Delta})$$

FULL certainty (we are sure $x = \hat{x}$)

The Gaussian Representation

$$p_x(x) = c \cdot e^{-\frac{(x - \hat{x})^2}{2 \cdot \sigma^2}}$$

It has two parameters

$\hat{x}$: expected value

σ^2 : variance

$$\hat{x} = \int_{-\infty}^{+\infty} x \cdot p_x(x) \cdot dx$$

$$\sigma^2 = \int_{-\infty}^{+\infty} (x - \hat{x})^2 \cdot p_x(x) \cdot dx$$

Multivariate Gaussians

2D case

$$p_{\mathbf{X}}(\mathbf{X}) = c \cdot e^{-\frac{1}{2}(\mathbf{X}-\hat{\mathbf{X}})^T \cdot \mathbf{P}^{-1} \cdot (\mathbf{X}-\hat{\mathbf{X}})}, \quad c = \frac{1}{\sqrt{(2 \cdot \pi)^n \cdot |\mathbf{P}|}}, n = 2$$

$$p_{\mathbf{X}}(.) : \mathbb{R}^2 \rightarrow \mathbb{R}^1$$

$$\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \hat{\mathbf{X}} = \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix}, \quad \mathbf{P} = \begin{bmatrix} P_{1,1} & P_{1,2} \\ P_{2,1} & P_{2,2} \end{bmatrix}$$

(covariance matrix)

$$p_{\mathbf{X}}(\mathbf{X}) = c \cdot e^{-\frac{1}{2} \cdot (\mathbf{X} - \hat{\mathbf{X}})^T \cdot \mathbf{P}^{-1} \cdot (\mathbf{X} - \hat{\mathbf{X}})}$$

in which

$$(\mathbf{X} - \hat{\mathbf{X}}) : \text{vector, } 2 \times 1$$

$$(\mathbf{X} - \hat{\mathbf{X}})^T : \text{vector, } 1 \times 2$$

$$\mathbf{P} : \text{matrix } 2 \times 2$$

$$p_{\mathbf{X}}(\mathbf{X}) : \text{scalar}$$

$$p_{\mathbf{X}}(\mathbf{X}) = c \cdot e^{-\frac{1}{2} \cdot (\mathbf{X} - \hat{\mathbf{X}})^T \cdot \mathbf{P}^{-1} \cdot (\mathbf{X} - \hat{\mathbf{X}})}$$

in which

$$(\mathbf{X} - \hat{\mathbf{X}}) : \text{vector, } n \times 1$$

$$(\mathbf{X} - \hat{\mathbf{X}})^T : \text{vector, } 1 \times n$$

$$\mathbf{P} : \text{matrix } n \times n$$

$$p_{\mathbf{X}}(\mathbf{X}) : \text{scalar}$$

The matrix $\mathbf{P}$ has special characteristics such as:

- * It is a symmetric matrix
- * It is a positive semi-definite matrix

Just these two parameters, **expected value** (a vector) and the **covariance matrix** fully define the PDF

So, we usually talk about these two parameters, implicitly knowing we are talking about a PDF.

Higher dimension cases?

Yes, and some of them of very high dimension
(thousands, or hundreds of thousands)
(not in our interest now)

We will try cases of 3 , 4 and 5D, in MTRN4010.

3D: 2D : pose!

4D: 2D Pose + longitudinal velocity

5D: 2D pose + speed + gyro bias

Higher dimension cases?

Flying vehicle:

9D (3D position + 3D attitude + 3D velocity vector)

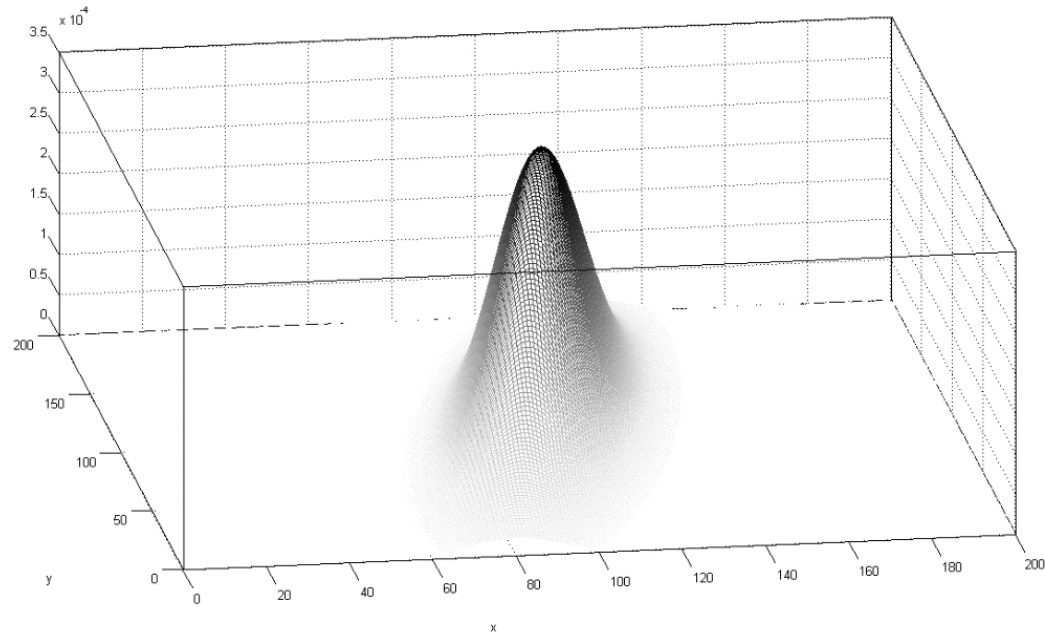
Covariance matrix : 9x9

Estimating state of system whose model is a PDE,
discretized.:1,000, 10,000...

How do they look?

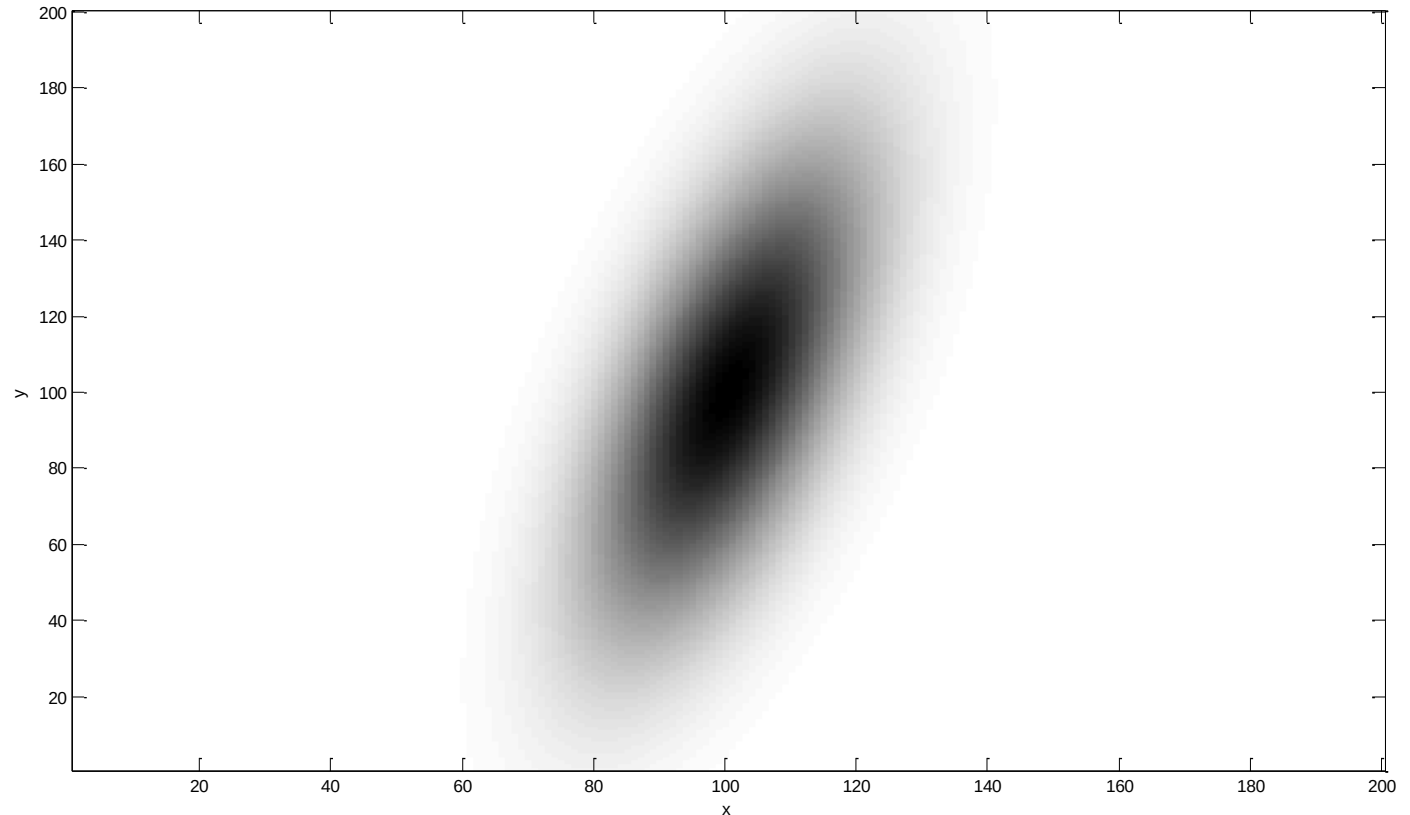
We can visualize 2D PDFs,

$$\hat{\mathbf{X}} = \begin{bmatrix} 100 \\ 100 \end{bmatrix}, \quad \mathbf{P}_X = \begin{bmatrix} 400 & 752 \\ 752 & 3600 \end{bmatrix}$$



How do they look? / We can visualize 2D PDFs,

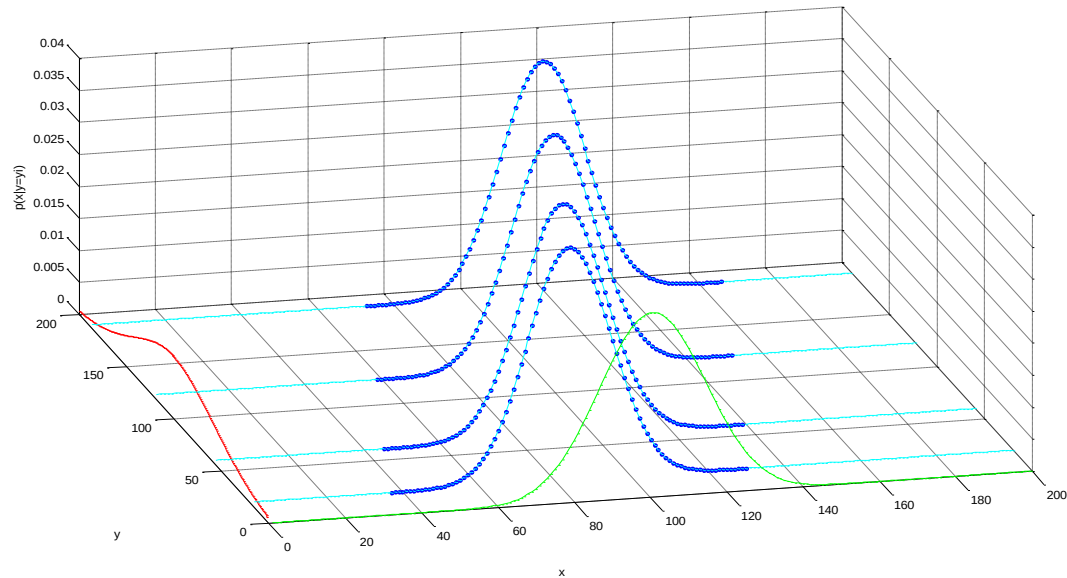
$$\hat{\mathbf{X}} = \begin{bmatrix} 100 \\ 100 \end{bmatrix}, \quad \mathbf{P}_x = \begin{bmatrix} 400 & 752 \\ 752 & 3600 \end{bmatrix}$$



Here I use grayscale : Dark regions correspond to high probability density

How do they look? / We can visualize 2D PDFs,

$$\hat{\mathbf{X}} = \begin{bmatrix} 100 \\ 100 \end{bmatrix}, \quad \mathbf{P}_x = \begin{bmatrix} 400 & 752 \\ 752 & 3600 \end{bmatrix}$$



Marginal PDFs, and some conditional ones.

Gaussian PDFs have properties which make them very efficient to be processed.

We will see an approach which exploits it.

(this is why we needed to refresh these concepts)

Next step in our work will be next week:

Bayesian Estimation

Gaussian Estimators.

KF / EKF (Kalman Filter / Extended Kalman Filter)